

ICT702 Data Visualisation

Visualising Variability

Module 5





I would like to acknowledge the traditional custodians of the
land we are meeting on,
The Gubbi Gubbi/Kabi Kabi peoples,
and pay my respect to Elders past, present and emerging.



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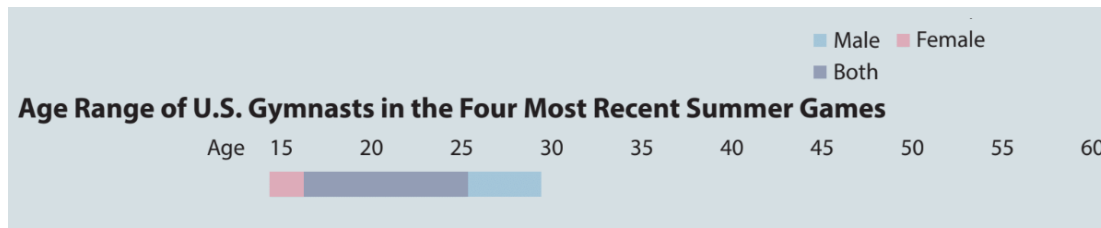
Learning Objectives for Module 5

- Create and interpret charts used to visualize the frequency distribution of a categorical variable
- Create and interpret histograms and frequency polygons, charts used to visualize the distribution of a quantitative variable
- Create and interpret visualizations comparing the distributions of two or more variables
- Create and interpret strip charts, recognize situations to use them, and employ techniques to improve their clarity
- Describe basic statistical measures of central location, variability, and distribution shape
- Create and interpret a box and whisker chart
- Create and interpret visualizations that depict the uncertainty resulting from sampling error
- Create and interpret charts that depict the uncertainty in predictions from simple regression models and time series models

Part 1

Data Visualization Makeover

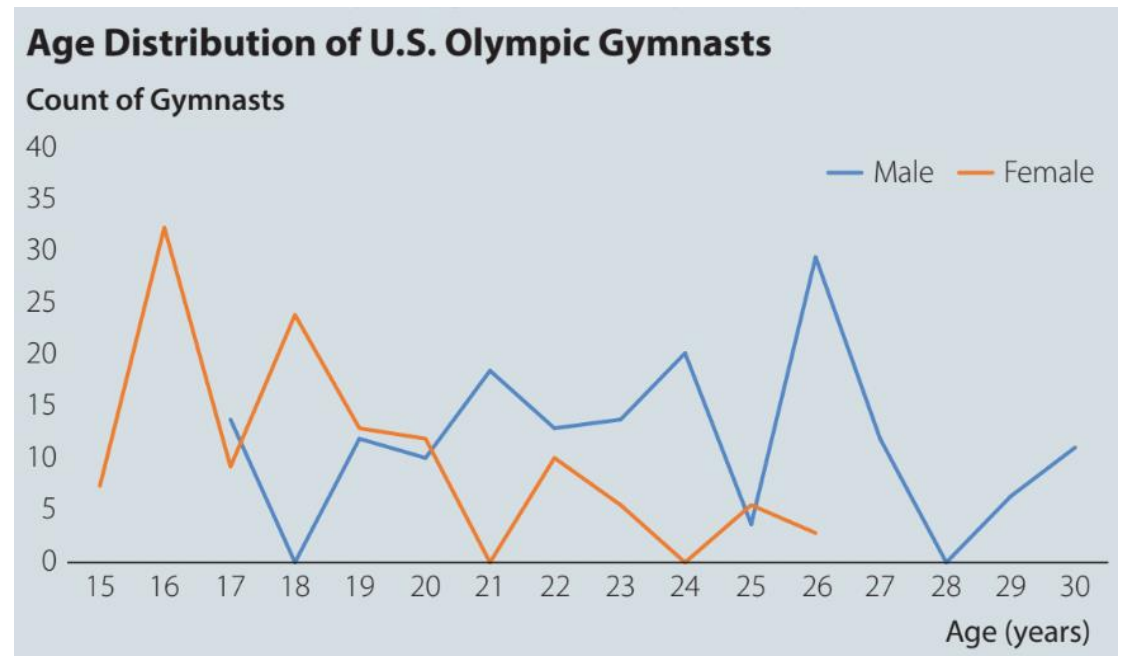
Overlapping range bar chart for ages of U.S. Olympic gymnasts



The third color in the range bar chart above is not intuitive and increases cognitive load.

The frequency polygons to the right reduce cognitive load and show the age distribution of male and female gymnasts.

Frequency polygon chart for U.S. Olympic gymnasts



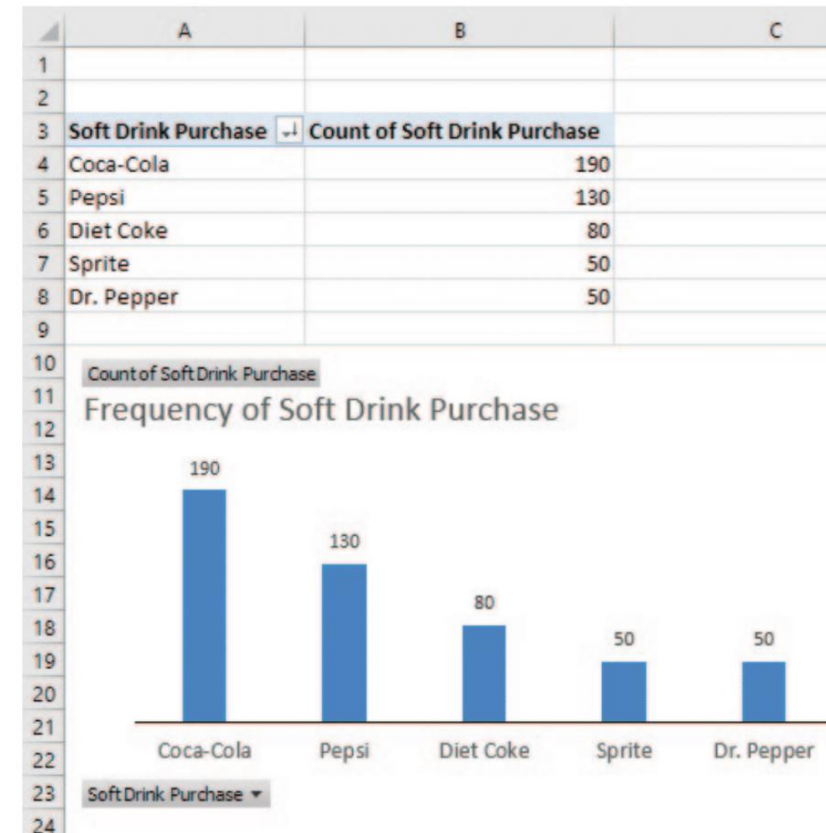
Creating Distributions from Data: Definitions

- A **random variable** is a quantity with values not known with certainty.
- **Variation** is the difference in a variable measured over observations.
- A **frequency distribution** describes the values of a variable and how often they appear in the data.
 - For a **categorical variable**, data consist of labels or names for which arithmetical manipulation is impossible.
 - For a **quantitative variable**, data consist of numerical values for which arithmetical manipulation is possible.
- A **sample** is a subset of the **population** that makes data collection feasible.

Visualise a Frequency Distribution: Column Chart and Pivot Table

	A
1	Soft Drink Purchase
2	Coca-Cola
3	Diet Coke
4	Pepsi
5	Diet Coke
6	Coca-Cola
7	Coca-Cola
8	Dr. Pepper
9	Diet Coke
10	Pepsi
11	Pepsi

PivotTable and PivotChart - frequency distribution of soft drink purchase data



Build a Relative Frequency Distribution

Create a frequency distribution for categorical data using the COUNTIF function

	A	B	C	D	E
1	Soft Drink Purchase		Bin	Frequency	Percent Frequency
2	Coca-Cola		Coca-Cola	=COUNTIF(A:A,C2)	=D2/SUM(\$D\$2:\$D\$6)
3	Diet Coke		Diet Coke	=COUNTIF(A:A,C3)	=D3/SUM(\$D\$2:\$D\$6)
4	Pepsi		Dr. Pepper	=COUNTIF(A:A,C4)	=D4/SUM(\$D\$2:\$D\$6)
5	Diet Coke		Pepsi	=COUNTIF(A:A,C5)	=D5/SUM(\$D\$2:\$D\$6)
6	Coca-Cola		Sprite	=COUNTIF(A:A,C6)	=D6/SUM(\$D\$2:\$D\$6)
7	Coca-Cola				

	A	B	C	D	E
1	Soft Drink Purchase		Bin	Frequency	Percent Frequency
2	Coca-Cola		Coca-Cola	190	38%
3	Diet Coke		Diet Coke	80	16%
4	Pepsi		Dr. Pepper	50	10%
5	Diet Coke		Pepsi	130	26%
6	Coca-Cola		Sprite	50	10%
7	Coca-Cola				

Relative Frequency and Percent Frequency

Definitions

The **relative frequency** of a bin equals the proportion of items belonging to a class:

$$\text{Relative Freq. of a bin} = \frac{\text{Freq. of a bin}}{n}$$

The **percent frequency** of a bin is the relative frequency multiplied by 100.

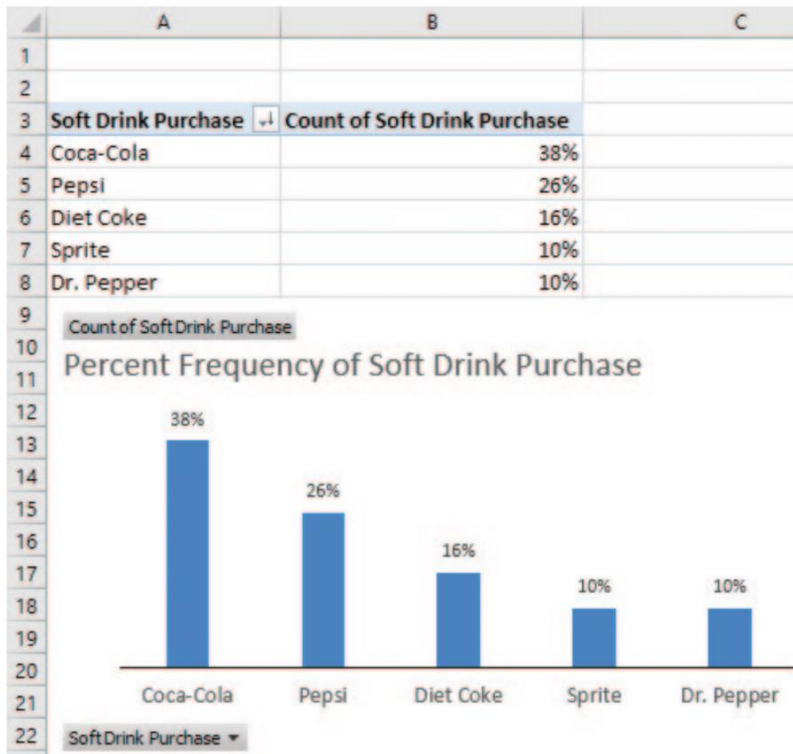
$$\text{Percent Freq. of a bin} = 100 \cdot \text{Relative Freq. of a bin}$$

A **probability distribution** characterizes the variability of a random variable.

- A percent frequency distribution estimates a probability distribution

Application of Benford's Law

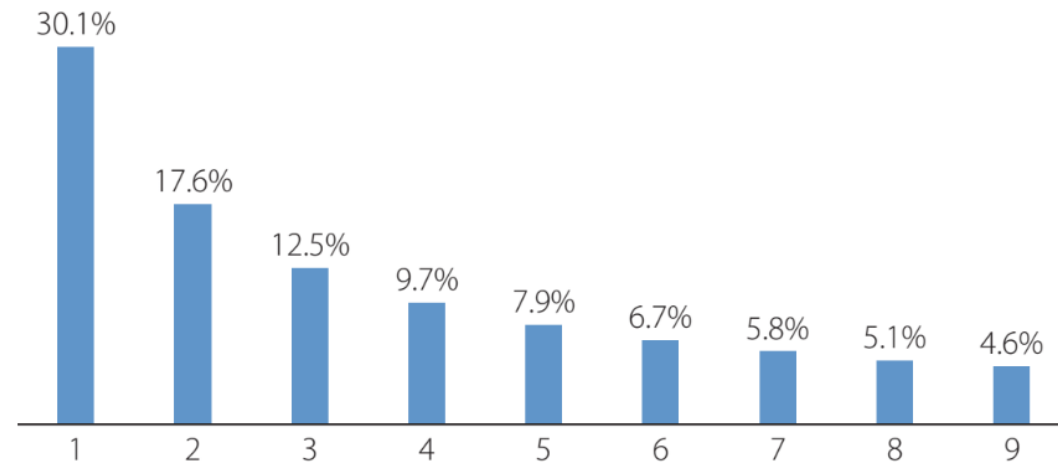
Percent frequency distribution of soft drink purchase data



Relative frequency distribution of data obeying Benford's Law

Benford's Law:

Proportion of First-Digit Observations



Benford's Law states that in many data sets, the proportion of observations in which the first digit is 1, 2, 3, 4, 5, 6, 7, 8, or 9, respectively, follows the distribution shown to the right.

End of Part 1